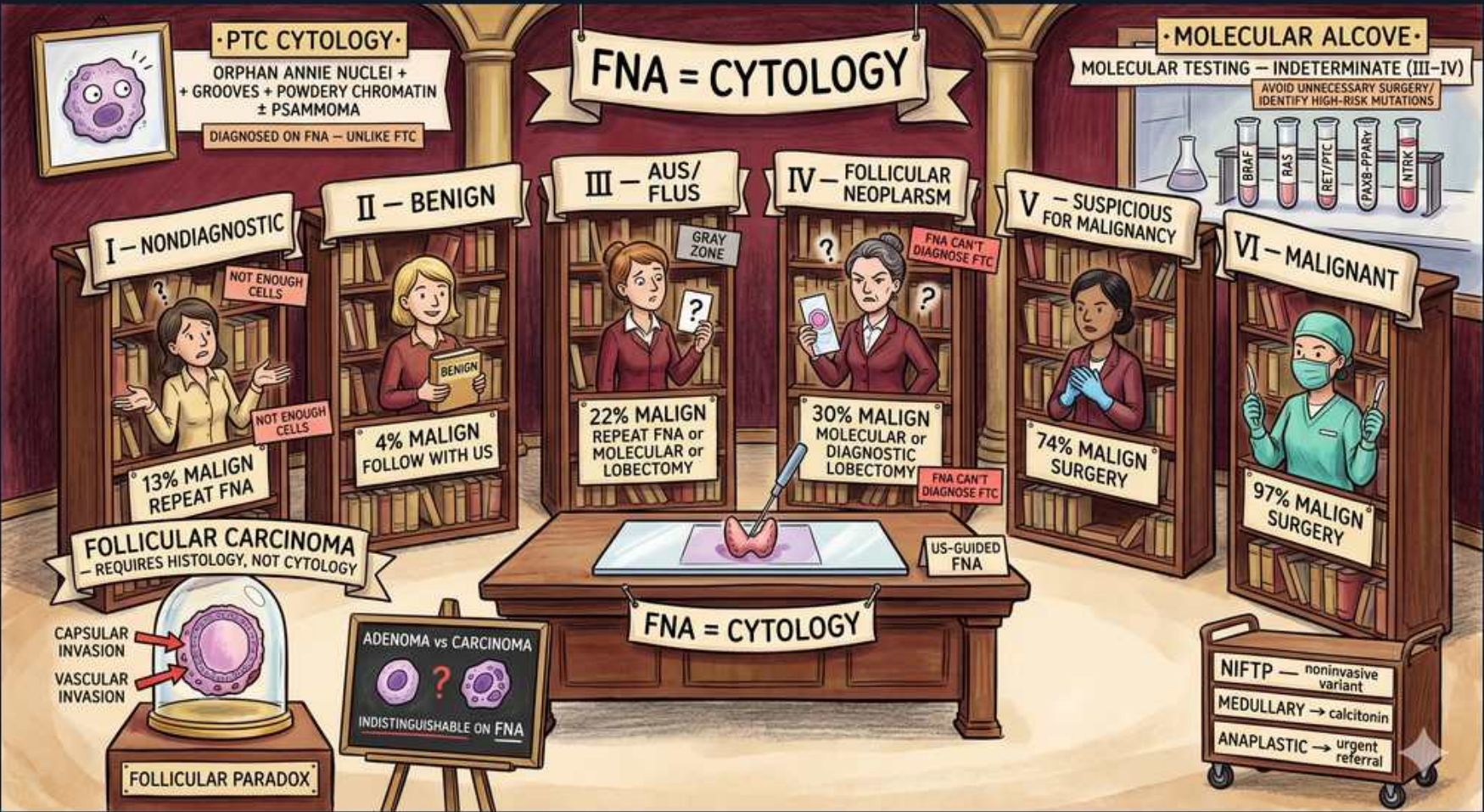


Thyroid — Bethesda Cytology & FNA



1. FNA = cytology

WHAT YOU SEE

Banner 'FNA = CYTOLOGY'

WHAT IT ENCODES

Fine-needle aspiration samples loose CELLS (cytology), not tissue architecture — it sees nuclear features but cannot show whether a tumor has broken through its capsule or invaded vessels. This is the single concept that unlocks the whole scene: FNA brilliantly diagnoses papillary cancer (a nuclear diagnosis) but is blind to follicular carcinoma (an architectural diagnosis). FNA is the first-line test for any sonographically suspicious nodule.

BOARD TRAP

Expecting FNA to diagnose or exclude follicular carcinoma — wrong because malignancy in follicular tumors is defined by capsular/vascular invasion seen only on histology, which cytology cannot assess.

2. I — Nondiagnostic

WHAT YOU SEE

Shelf I 'NOT ENOUGH CELLS, 13% MALIGN, REPEAT FNA'

WHAT IT ENCODES

Bethesda I = nondiagnostic/unsatisfactory (cyst fluid only or too few follicular cells). Risk of malignancy is low but not zero (~5–13%). Action: REPEAT FNA under ultrasound guidance, ideally with on-site adequacy assessment; persistently nondiagnostic + suspicious sonography → consider surgery.

BOARD TRAP

Calling a nondiagnostic specimen 'benign' and stopping — wrong; an inadequate sample tells you nothing and must be repeated.

3. II — Benign

WHAT YOU SEE

Shelf II 'BENIGN, ~4% MALIGN, FOLLOW WITH US'

WHAT IT ENCODES

Bethesda II = benign (colloid nodule, lymphocytic/Hashimoto thyroiditis). Malignancy risk is the lowest at ~0–4%. Action: no surgery — ultrasound surveillance. This category covers the majority of FNAs.

BOARD TRAP

Operating on or over-working-up a benign nodule — wrong; the ~4% risk warrants follow-up US, not resection.

4. III — AUS/FLUS

WHAT YOU SEE

Shelf III 'AUS/FLUS, ~22% MALIGN, GRAY ZONE, REPEAT FNA or MOLECULAR or LOBECTOMY'

WHAT IT ENCODES

Bethesda III = atypia/follicular lesion of undetermined significance — the gray zone (~6–18%, often quoted ~22%). Action: repeat FNA, OR molecular testing to refine risk, OR diagnostic lobectomy. Molecular testing is most useful here to avoid unnecessary surgery.

BOARD TRAP

Jumping straight to total thyroidectomy for indeterminate cytology — wrong; first refine with repeat FNA/molecular testing, then lobectomy if still indeterminate.

5. IV — Follicular neoplasm

WHAT YOU SEE

Shelf IV 'FOLLICULAR NEOPLASM, ~30% MALIGN, FNA CAN'T DIAGNOSE FTC, MOLECULAR or DIAGNOSTIC LOBECTOMY'

WHAT IT ENCODES

Bethesda IV = follicular neoplasm / suspicious for follicular neoplasm (~10–40%, often quoted ~30%). This IS the follicular paradox category — cytology cannot separate adenoma from carcinoma. Action: molecular testing OR diagnostic (hemi-)lobectomy to obtain histology and look for capsular/vascular invasion.

BOARD TRAP

Reporting 'follicular carcinoma' off the FNA — wrong; FNA cannot make that call, so the diagnostic step is lobectomy for histology (or molecular testing to triage).

6. V — Suspicious for malignancy

WHAT YOU SEE

Shelf V 'SUSPICIOUS, 74% MALIGN, SURGERY'

WHAT IT ENCODES

Bethesda V = suspicious for malignancy (features present but not definitive, usually suspicious for papillary). Malignancy risk ~60–75%. Action: surgery — lobectomy or total thyroidectomy depending on size/risk.

BOARD TRAP

Treating 'suspicious' as a benign deferral — wrong; with ~74% malignancy risk this category proceeds to surgery.

7. VI — Malignant

WHAT YOU SEE

Shelf VI 'MALIGNANT, 97% MALIGN, SURGERY'

WHAT IT ENCODES

Bethesda VI = malignant (diagnostic cytology, almost always papillary). Malignancy risk ~97–99%. Action: definitive surgery (total thyroidectomy for most; lobectomy for small low-risk PTC).

BOARD TRAP

Repeating the FNA on a frankly malignant result instead of referring for surgery.

8. Follicular paradox

WHAT YOU SEE

'FOLLICULAR CARCINOMA — REQUIRES HISTOLOGY, NOT CYTOLOGY'

WHAT IT ENCODES

The follicular paradox: follicular adenoma and follicular carcinoma have IDENTICAL cytology — the only difference is capsular and/or vascular (angioinvasion) invasion, which is a tissue-architecture feature invisible to a cytology smear. Therefore the diagnosis of follicular carcinoma can only be made on the resected specimen (lobectomy), never on FNA. The same logic applies to the follicular variant of Hürthle cell carcinoma.

BOARD TRAP

Believing a 'benign' or 'follicular neoplasm' FNA rules out cancer — wrong; up to ~15–30% of follicular neoplasms are carcinoma, only provable on histology.

9. Capsular/vascular invasion

WHAT YOU SEE

'CAPSULAR INVASION / VASCULAR INVASION'

WHAT IT ENCODES

Malignancy in a follicular-patterned tumor is DEFINED histologically by capsular invasion (tumor through the full thickness of the capsule) and/or vascular invasion (tumor in capsular/extracapsular vessels). These criteria require an intact resected capsule — impossible to judge on aspirated cells — which is exactly why FNA cannot diagnose FTC.

BOARD TRAP

Assuming nuclear atypia on FNA equals follicular carcinoma — invasion (not atypia) defines follicular malignancy, and invasion needs histology.

10. Adenoma vs carcinoma identical on FNA

WHAT YOU SEE

'ADENOMA vs CARCINOMA — INDISTINGUISHABLE ON FNA'

WHAT IT ENCODES

Follicular adenoma and follicular carcinoma are cytologically indistinguishable because both are uniform microfollicular proliferations; the malignant one simply has invasion you cannot see on a smear. This is the mechanistic reason Bethesda IV exists as 'neoplasm' rather than a benign/malignant call.

BOARD TRAP

Distinguishing adenoma from carcinoma on cytology grounds — impossible; only the histology of the capsule/vessels separates them.

11. PTC nuclear features on FNA

WHAT YOU SEE

'PTC CYTOLOGY: ORPHAN ANNIE NUCLEI, GROOVES, POWDERY CHROMATIN, PSAMMOMA — UNLIKE FTC'

WHAT IT ENCODES

Papillary thyroid carcinoma IS a cytologic (nuclear) diagnosis — FNA reliably catches it via Orphan-Annie-eye nuclei (empty/clear with intranuclear pseudoinclusions), nuclear grooves, powdery/finely dispersed chromatin, and psammoma bodies (laminated calcifications). Mnemonic 'PAPILLARY = nuclear PAPER': Pseudoinclusions, Annie eyes, Psammoma, grooves. This is the deliberate contrast with the architecture-dependent follicular cancer.

BOARD TRAP

Saying papillary carcinoma needs histology like follicular does — wrong; PTC is defined by nuclear features, so FNA can diagnose it directly (Bethesda VI).

12. Molecular testing for III/IV

WHAT YOU SEE

'MOLECULAR TESTING — INDETERMINATE (III–IV), AVOID UNNECESSARY SURGERY, IDENTIFY HIGH-RISK MUTATIONS'

WHAT IT ENCODES

Molecular testing (e.g., Afirma GSC, ThyroSeq) is applied to indeterminate cytology (Bethesda III and IV) to reclassify nodules: a 'rule-out' (high negative predictive value) panel lets benign-behaving nodules avoid surgery, while detected high-risk mutations (BRAF V600E, RAS, RET/PTC, PAX8/PPAR γ , TERT promoter) raise concern and favor resection. BRAF V600E in particular is highly specific for papillary carcinoma.

BOARD TRAP

Using molecular testing on already-benign (II) or already-malignant (VI) nodules — wrong; its value is specifically in the indeterminate III/IV gray zone to triage who needs surgery.

13. US-guided FNA

WHAT YOU SEE

'US-GUIDED FNA'

WHAT IT ENCODES

FNA is performed under real-time ultrasound guidance to sample the most suspicious portion of the nodule (microcalcifications, irregular margins, taller-than-wide, hypoechoic, increased vascularity), which maximizes adequacy and minimizes nondiagnostic (Bethesda I) results. Nodule selection for FNA is risk-stratified by sonographic pattern and size (e.g., TI-RADS / ATA categories).

BOARD TRAP

FNA-ing every nodule regardless of sonographic risk — wrong; small purely-cystic/spongiform nodules below size cutoffs are followed, not aspirated.

14. NIFTP / MTC / anaplastic notes

WHAT YOU SEE

'NIFTP NONINVASIVE VARIANT; MEDULLARY = CALCITONIN; ANAPLASTIC URGENT REFERRAL'

WHAT IT ENCODES

Special cases on cytology: NIFTP (non-invasive follicular thyroid neoplasm with papillary-like nuclear features) is an indolent, reclassified-as-non-cancer entity treated by lobectomy alone. Medullary carcinoma cytology should prompt serum calcitonin (and CEA) plus RET/MEN2 evaluation. Anaplastic carcinoma on FNA (older patient, rapidly enlarging mass) is an oncologic emergency needing urgent multidisciplinary referral.

BOARD TRAP

Treating NIFTP as aggressive papillary cancer (over-treatment) or, conversely, delaying work-up of an anaplastic/medullary cytology result — each has a distinct, time-sensitive pathway.